CALCULATED THAT MARS HAD AN ELLIPTICAL ORBIT AFTER **40 FAILED ATTEMPTS**

HIS **3RD LAW** TOOK HIM **10 YEARS TO FORMULATE**

DISCOVERED **SATURN** TAKES ROUGHLY **29 EARTH YEARS** TO ORBIT THE SUN

TYCHO **BRAHE**

Danish astronomer Tycho Brahe accurately cataloged the stars before the telescope.

In 1577, Brahe (1546–1601) observed a new star in the constellation of Cassiopeia and determined to accurately record celestial bodies. He built a set of instruments with which to measure the positions of the stars and spent 20 years compiling data. The accuracy of his observations surpassed those of any other practicing astronomer and enabled his assistant, Johannes Kepler, to calculate his laws of planetary motion.

the planets and asked Kepler, who had the better mathematical skills, to investigate the complicated orbit of Mars.

Laws of planetary motion

When Brahe died in 1601, Kepler took over his role as Imperial Mathematician to Rudolph II, Holy Roman Emperor. Using Brahe's extensive observations, compiled over 20 years, Kepler resumed his study of Mars, which resulted in his three laws of planetary motion.

Unable to match Brahe's observational data to Copernicus's criteria—that each planet has a circular orbit and moves at a constant velocity—Kepler adjusted his calculations. He began experimenting with different orbital shapes, eventually concluding that a planet's orbit must be elliptical (like a stretched circle) and that the extent (or eccentricity) of that ellipse would vary from planet to planet. This was Kepler's first law of planetary motion and successfully explained the apparent retrograde (backward) motion of Mars as viewed from Earth.

His second law stated that a planet's velocity was not constant and that the closer it came to the Sun, the faster it would travel. These laws, published in his 1609 text *Astronomia Nova*, represented a landmark in astronomical history.

Published a decade later, Kepler's third law of motion mathematically described the relationship between a planet's year (orbital period) and its average distance from the Sun. He calculated the orbital period for each planet and in comparing them realized that if a planet is twice as far from the Sun as another planet, then its orbital period will be three times longer than that of the nearer planet.

Influences and achievements

The accuracy of Kepler's laws gave credence to the heliocentric view of the Universe and ensured its wide acceptance. His work in astronomy gave Newton the basis on which he would later devise his universal laws of gravitation and marked a profound shift in scientific thought.

While he is best known for his astronomical work, Kepler also made great advances in the study of optics and produced one of the first key texts in this field. With achievements that included explaining the visual mechanics of depth perception and proposing designs for eyewear to combat near- and farsightedness, he is also considered a pioneer in the field of modern optics.